

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Botany**Paper: BOB-502 — Comparative Studies of Phanerogams**

Time: Three Hours

Full Marks: 70

Note: Attempt five questions in all, selecting two questions from each Section-A and B. Question No. 9 (Section-C) is compulsory. The figures in the right-hand margin indicate marks.

SECTION – A

Question 1. Give a detailed account of the order Gnetales with special reference to *Gnetum*.
[15]

Question 2. Discuss briefly any two of the following: [7.5 × 2 = 15]

- (a) Embryogeny in conifers
- (b) *Pentoxylon*
- (c) Origin of Gymnosperms

Question 3. With the help of suitable diagrams, describe briefly the morphology and reproduction in *Zamia*. [15]

Question 4. With the help of labelled diagrams, differentiate between any three of the following: [5 × 3 = 15]

- (a) Male and female strobili of *Ginkgo*
- (b) Anatomy of leaves of *Gnetum* and *Zamia*
- (c) Manoxylic and pycnoxylic wood
- (d) Female cones of *Zamia* and *Biota*

SECTION – B

Question 5. With the help of floral formula and floral diagram, describe the family Scrophulariaceae. Give names of two economically important plants of this family. [15]

Question 6. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Anther culture
- (b) Outline of Hutchinson's system of classification
- (c) Numerical taxonomy

Question 7. Write salient features of the family Rutaceae and discuss its economic importance. [15]

Question 8. Differentiate between any three of the following pairs of families: $[5 \times 3 = 15]$

- (a) Scrophulariaceae and Meliaceae
- (b) Annonaceae and Rutaceae
- (c) Moraceae and Liliaceae
- (d) Asteraceae and Convolvulaceae

SECTION – C

Question 9. Answer the following in not more than 50 words each: $[1 \times 10 = 10]$

- (a) Differentiate between haplocheilic and syndetocheilic stomata.
- (b) What is a feeder and where is it found?
- (c) Differentiate between embryogeny of *Gnetum* and *Biota*.
- (d) Why are Cycadales and Ginkgoales called living fossils?
- (e) What is a seed scale complex?
- (f) Name a family in which obdiplostemonous condition is found.
- (g) Give the botanical name of a medicinal plant of Convolvulaceae.
- (h) Name a family in which basal placentation is found.
- (i) Write the number of stamens found in a male flower of *Euphorbia*.
- (j) Define hypogynous, epigynous and perigynous flowers.

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